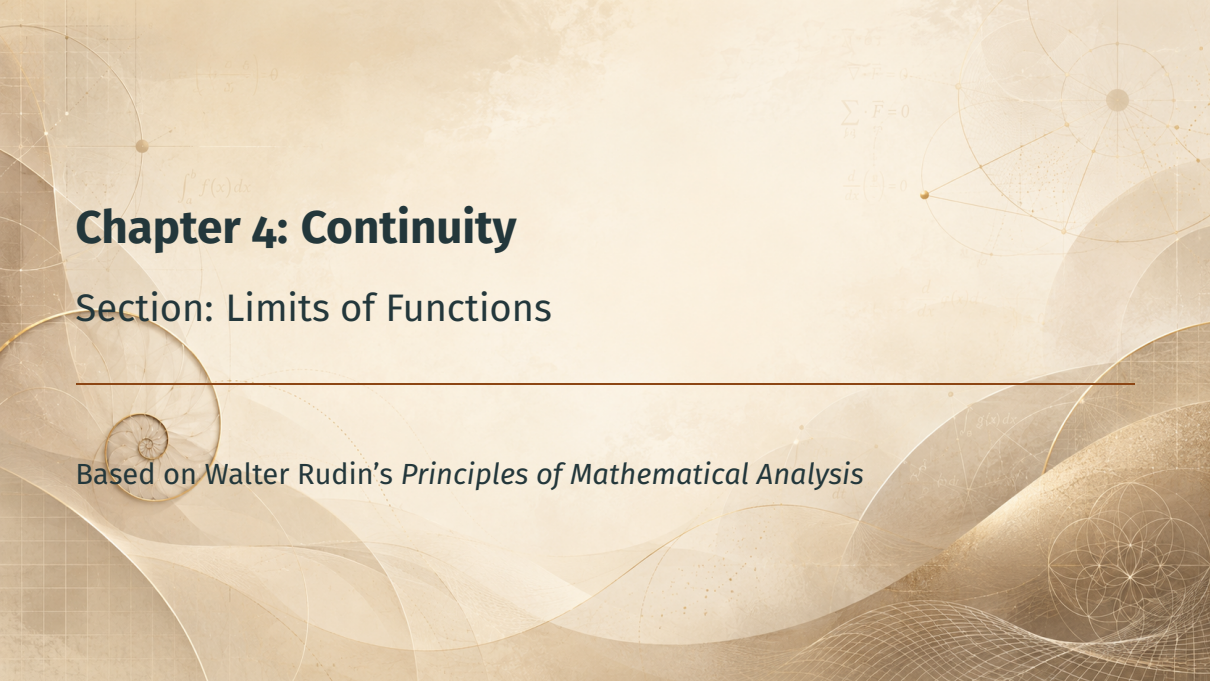


The background is a light beige color with various mathematical and geometric elements. In the top left, there is a formula $\left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) \cdot \theta$. Below it is $\int_a^b f(x) dx$. In the top right, there are formulas $\nabla \cdot \vec{F} = 0$, $\sum_{i=1}^n \vec{F}_i = 0$, and $\frac{d}{dx} \left(\frac{g}{f} \right) = 0$. On the right side, there is a diagram of a sphere with lines connecting points on its surface. In the bottom right, there is a diagram of a circle with internal lines forming a star-like pattern. A horizontal line is drawn across the middle of the page.

Chapter 4: Continuity

Section: Limits of Functions

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

The Definition of a Limit

Limits via Sequences

The Algebra of Functions

Summary

Motivation

From Sequences to Functions

- Chapter 3: limits of **sequences** — $\lim_{n \rightarrow \infty} p_n = p$

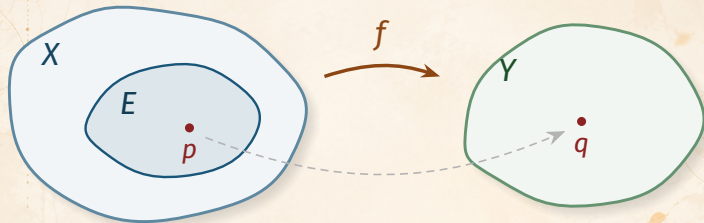
From Sequences to Functions

- Chapter 3: limits of **sequences** — $\lim_{n \rightarrow \infty} p_n = p$
- Now: limits of **functions** — $\lim_{x \rightarrow p} f(x) = q$

From Sequences to Functions

- Chapter 3: limits of **sequences** — $\lim_{n \rightarrow \infty} p_n = p$
- Now: limits of **functions** — $\lim_{x \rightarrow p} f(x) = q$
- Bridge: Theorem 4.2 reduces function limits to **sequence limits**

The General Setting



Setting: f maps $E \subset X$ into Y , where X, Y are **metric spaces**.

Why so general? Real, complex, and vector-valued functions are all special cases — the general proofs are *no harder*.

$$\nabla \cdot F = 0$$

$$\frac{d}{dx}(e^x) = e^x$$

$$\int_a^b f(x) dx$$

$$e^{i\pi} + 1 = 0$$

The Definition of a Limit

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Definition 4.1: Limit of a Function

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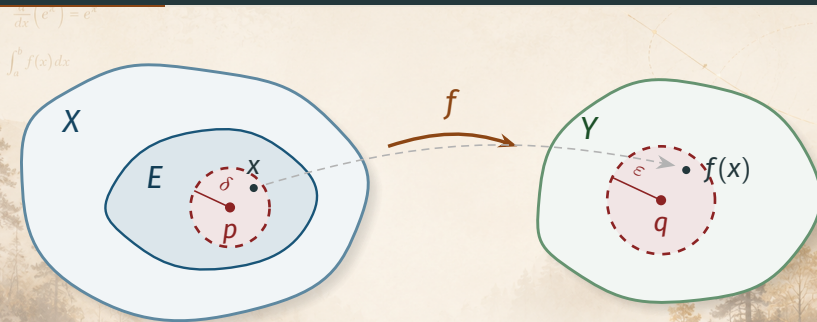
if there is a point $q \in Y$ with the following property: for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$d_Y(f(x), q) < \varepsilon \quad (2)$$

for all points $x \in E$ for which

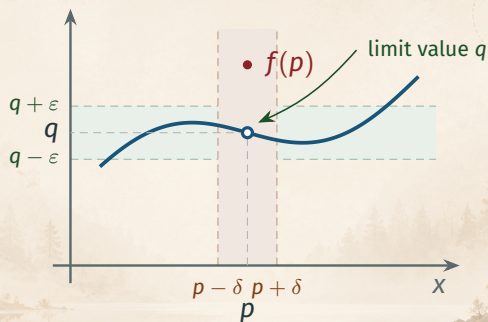
$$0 < d_X(x, p) < \delta. \quad (3)$$

The Picture in Metric Spaces



Reading the definition: every point $x \neq p$ inside the δ -ball (and in E) is sent by f into the ε -ball around q .

A Limit Where $f(p) \neq q$



p need **not** be in E ; even if $p \in E$, possibly $f(p) \neq \lim_{x \rightarrow p} f(x)$.

Remarks on Definition 4.1

Remarks

- d_X, d_Y are the distances in X, Y . For $\mathbb{R}, \mathbb{C}, \mathbb{R}^k$: absolute values or norms of differences.

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- d_X, d_Y are the distances in X, Y . For $\mathbb{R}, \mathbb{C}, \mathbb{R}^k$: absolute values or norms of differences.
- $p \in X$ must be a limit point of E , but p **need not belong to E** .
- Even if $p \in E$, possibly $f(p) \neq \lim_{x \rightarrow p} f(x)$: condition (3) **excludes $x = p$** .

$$e^{i\pi} + 1 = 0$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$y = \frac{\sin x}{x}$$

Limits via Sequences

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Theorem 4.2: Sequential Characterization

Theorem 4.2

Let X, Y, E, f , and p be as in Definition 4.1. Then

$$\lim_{x \rightarrow p} f(x) = q \quad (4)$$

if and only if

$$\lim_{n \rightarrow \infty} f(p_n) = q \quad (5)$$

for **every** sequence $\{p_n\}$ in E such that

$$p_n \neq p, \quad \lim_{n \rightarrow \infty} p_n = p. \quad (6)$$

Proof Strategy: Theorem 4.2

Goal: Prove the equivalence $(4) \iff (5)$.

1. $(\Rightarrow)$ Assume (4). Chain the two definitions: $\varepsilon \rightsquigarrow \delta \rightsquigarrow N$.
2. $(\Leftarrow)$ Prove the contrapositive: if (4) **fails**, build a **bad sequence** with $\delta_n = 1/n$.

Technique: the second half is a classic **sequence-construction**.

Proof of Theorem 4.2 — ($\Rightarrow$)

Step 1: From ε to δ

Suppose (4) holds. Choose $\{p_n\}$ in E satisfying (6). Let $\varepsilon > 0$ be given. Then there exists $\delta > 0$ such that

$$d_Y(f(x), q) < \varepsilon \quad \text{if } x \in E \text{ and } 0 < d_X(x, p) < \delta.$$

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Step 2: From δ to N

Since $p_n \rightarrow p$ and $p_n \neq p$, there exists N such that

$$n > N \implies 0 < d_X(p_n, p) < \delta \implies d_Y(f(p_n), q) < \varepsilon.$$

Hence $f(p_n) \rightarrow q$, i.e., (5) holds.

■ (forward direction)

Proof of Theorem 4.2 — ($\Leftarrow$), by Contraposition

Step 1: Negate (4)

Suppose (4) is **false**. Then there exists some $\varepsilon > 0$ such that for **every** $\delta > 0$ there is a point $x \in E$ (depending on δ) with

$$d_Y(f(x), q) \geq \varepsilon \quad \text{but} \quad 0 < d_X(x, p) < \delta.$$

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Step 2: Build the bad sequence

Take $\delta_n = \frac{1}{n}$ ($n = 1, 2, 3, \dots$): we obtain points $p_n \in E$ with $0 < d_X(p_n, p) < \frac{1}{n}$ but $d_Y(f(p_n), q) \geq \varepsilon$.

The sequence $\{p_n\}$ satisfies (6), yet (5) is **false**. 

Corollary: Uniqueness of Limits

Corollary

If f has a limit at p , this limit is **unique**.

Why: Theorem 3.2(b) (limits of sequences are unique) + Theorem 4.2.

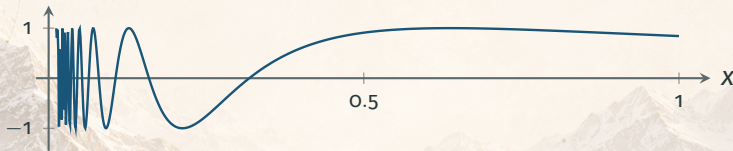
Recall: Theorem 3.2(b)

A sequence in a metric space converges to *at most one* point.

Example: A Limit That Fails to Exist

Example (the criterion in action)

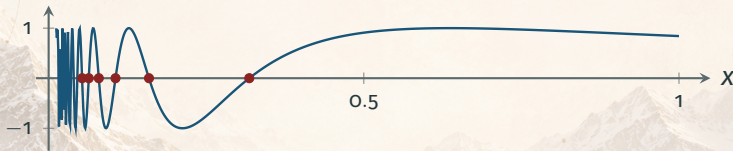
Does $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$ exist? Here $E = (0, 1]$; note $0 \notin E$, but 0 is a limit point of E .



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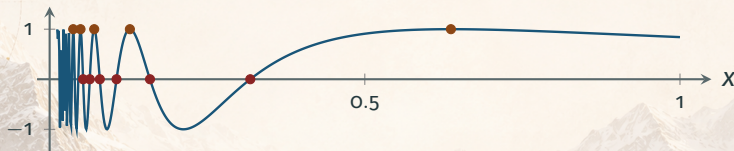


• Along $a_n = \frac{1}{n\pi} \rightarrow 0$: $f(a_n) = \sin(n\pi) = 0 \rightarrow 0$

Example: A Limit That Fails to Exist

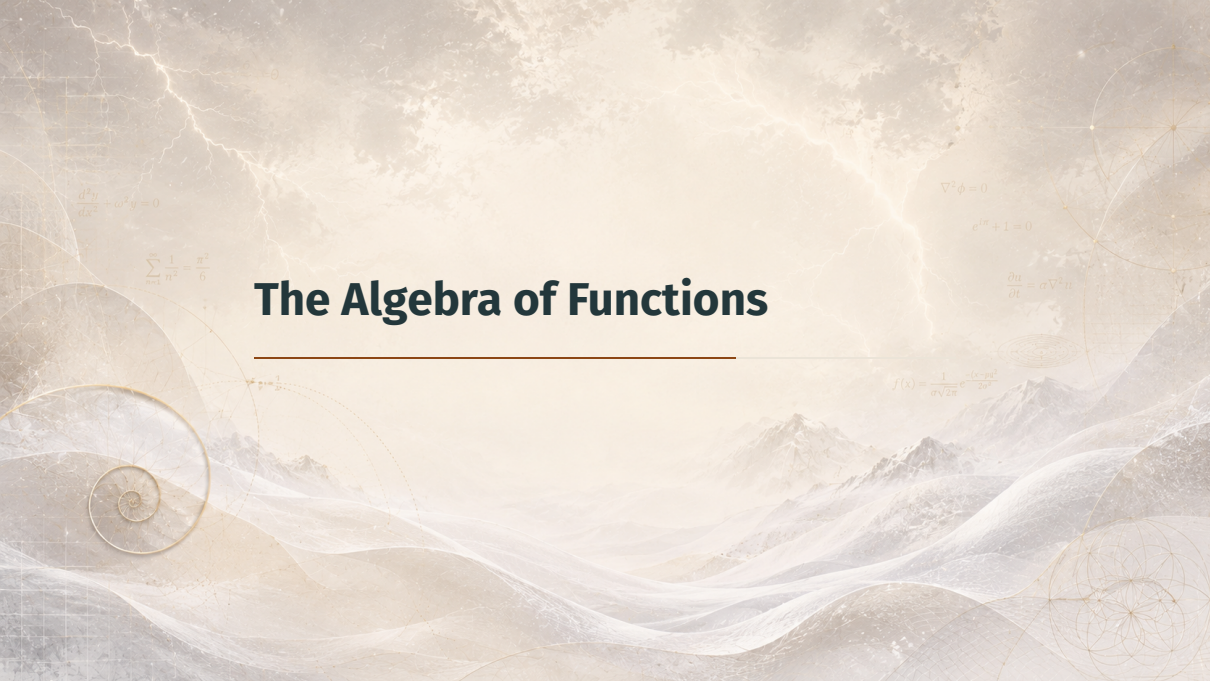
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- Along $a_n = \frac{1}{n\pi} \rightarrow 0$: $f(a_n) = \sin(n\pi) = 0 \rightarrow 0$
- Along $b_n = \frac{2}{(4n+1)\pi} \rightarrow 0$: $f(b_n) = \sin\left(2n\pi + \frac{\pi}{2}\right) = 1 \rightarrow 1$

Two sequences, two different limits $\implies$
by Theorem 4.2, the limit does **not exist**.

The background is a complex, artistic composition. It features a landscape with mountains and wavy, ethereal lines in shades of blue and white. Overlaid on this are various mathematical and geometric elements: a golden spiral on the left, a golden triangle at the top left, a golden rectangle at the top right, and a golden circle with internal lines on the right. Several mathematical formulas are scattered throughout the image.

The Algebra of Functions

$$\frac{d^2 y}{dx^2} + \omega^2 y = 0$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$\nabla^2 \phi = 0$$

$$e^{im} + 1 = 0$$

$$\frac{\partial u}{\partial t} = \sigma \nabla^2 u$$

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Definition 4.3: Operations on Functions

Definition 4.3 (complex functions)

Let $f, g: E \rightarrow \mathbb{C}$. Define, for each $x \in E$:

$$(f + g)(x) = f(x) + g(x), \quad (f - g)(x) = f(x) - g(x),$$

$$(fg)(x) = f(x)g(x), \quad \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \text{ (only where } g(x) \neq 0).$$

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Notation

- $f = c$: *constant function*, $f(x) = c$ for all $x \in E$.
- $f \geq g$ (real f, g): means $f(x) \geq g(x)$ for every $x \in E$.

Definition 4.3: Vector-Valued Functions

Definition 4.3 (vector-valued functions)

If $\mathbf{f}, \mathbf{g}: E \rightarrow \mathbb{R}^k$, define

$$(\mathbf{f} + \mathbf{g})(x) = \mathbf{f}(x) + \mathbf{g}(x), \quad (\mathbf{f} \cdot \mathbf{g})(x) = \mathbf{f}(x) \cdot \mathbf{g}(x),$$

and for $\lambda \in \mathbb{R}$, $(\lambda \mathbf{f})(x) = \lambda \mathbf{f}(x)$.

Note: $\mathbf{f} \cdot \mathbf{g}$ is the **inner product** — a scalar-valued function.

Theorem 4.4: Limit Laws

Theorem 4.4

Suppose $E \subset X$, a metric space, p is a limit point of E , f and g are complex functions on E , and

$$\lim_{x \rightarrow p} f(x) = A, \quad \lim_{x \rightarrow p} g(x) = B.$$

Then:

$$(a) \lim_{x \rightarrow p} (f + g)(x) = A + B;$$

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Then:

(a) $\lim_{x \rightarrow p} (f + g)(x) = A + B;$

(b) $\lim_{x \rightarrow p} (fg)(x) = AB;$

(c) $\lim_{x \rightarrow p} \left(\frac{f}{g} \right) (x) = \frac{A}{B},$ if $B \neq 0.$

Proof of Theorem 4.4

Proof

In view of Theorem 4.2, these assertions follow immediately from the analogous properties of **sequences** (Theorem 3.3). $e^{i\pi} + 1 = 0$ ■

The bridge in action:

(4) function limit $\xleftrightarrow{\text{Thm 4.2}}$ (5) sequence limits $\xleftrightarrow{\text{Thm 3.3}}$ limit laws

Remark: Vector-Valued Limit Laws

Remark

If $\mathbf{f}$ and $\mathbf{g}$ map E into $\mathbb{R}^k$, then (a) remains true, and (b) becomes:

$$(b') \lim_{x \rightarrow p} (\mathbf{f} \cdot \mathbf{g})(x) = \mathbf{A} \cdot \mathbf{B}.$$

(Compare Theorem 3.4.) There is no analogue of (c): we cannot divide by a vector.


$$e^{i\pi} + 1 = 0$$

$$\phi = \frac{1 + \sqrt{5}}{2}$$

Summary

$$\nabla^2 \psi + k^2 \psi = 0$$

$$\frac{d}{dx}(\sin x) = \cos x$$

Summary

Key takeaways from this section:

- **Definition 4.1:** $\lim_{x \rightarrow p} f(x) = q$ via ε - δ in metric spaces; p a limit point of E , value $f(p)$ irrelevant

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- **Definition 4.1:** $\lim_{x \rightarrow p} f(x) = q$ via ε - δ in metric spaces; p a limit point of E , value $f(p)$ *irrelevant*
- **Theorem 4.2:** function limits $\iff$ sequence limits; corollary: limits are *unique*

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- **Theorem 4.2:** function limits $\iff$ sequence limits; corollary: limits are *unique*
- **Theorem 4.4:** limit laws for $f + g, fg, f/g$ (and $\mathbf{f} + \mathbf{g}, \mathbf{f} \cdot \mathbf{g}$)

Coming up next:

Continuous Functions — what happens when $\lim_{x \rightarrow p} f(x) = f(p)$.